

# $\beta$ Derivation: Exact Decomposition and Remaining Gap

Formula Verified Exact;  $f_{II} = 1/n$  Hypothesis; Gap Located at  $H_{\text{after}}$

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## Abstract

The  $\beta$  transfer rate formula is verified exact from per-event simulation data:

$$\beta = \frac{1}{\kappa} \left[ \bar{F} \cdot (H_{\text{after}} \overline{\ln(m + \kappa)} - H_{\text{before}} \overline{\ln m}) + \text{Cov}(F, H_{\text{after}} \ln(m + \kappa)) - \text{Cov}(F, H_{\text{before}} \ln m) \right]$$

Numerical verification:  $\beta_{\text{formula}} = 2.2571$ ,  $\beta_{\text{actual}} = 2.2571$ . Residual = 0.

The remaining gap is entirely in  $H_{\text{after}}$ :  $H_{\text{after}}^{\text{derived}} = 1.836$  vs  $H_{\text{after}}^{\text{sim}} = 1.798$  (2.1% in  $H$ , 16% in  $\beta$ ).

A new hypothesis:  $f_{II} = 1/n$  (fraction of R-events with peaked neighbours equals  $1/n$ , derived from  $n$  equally-populated  $\Psi$ -modes from  $\text{SO}(3)$ ). This gives  $H_{\text{after}} = (1 - 1/n)H_{\text{uniform}} + (1/n)H_{\text{peaked}} = 1.836$ .

The remaining 16% gap in  $\beta$  is precisely the discrepancy between  $f_{II}^{\text{derived}} = 1/n = 0.125$  and  $f_{II}^{\text{measured}} = 0.147$ .

## 1 Formula Verified Exact

### Exact Decomposition from Simulation

Per-event computation from V20.3 (2016 events, seed 42):

$$\begin{aligned} \beta &= \frac{1}{\kappa} \mathbb{E} [F_i (H_{\text{after},i} \ln(m_i + \kappa) - H_{\text{before},i} \ln m_i)] \\ &= \frac{1}{\kappa} \left[ \bar{F} (\bar{H}_a \overline{\ln(m + \kappa)} - \bar{H}_b \overline{\ln m}) + \text{Cov}(F, H_a \ln(m + \kappa)) - \text{Cov}(F, H_b \ln m) \right] \end{aligned}$$

|                                             |               |
|---------------------------------------------|---------------|
| Main term $\bar{F}(\dots)/\kappa$           | 2.2548        |
| $\text{Cov}(F, H_a \ln(m + \kappa))/\kappa$ | -0.0395       |
| $\text{Cov}(F, H_b \ln m)/\kappa$           | -0.0418       |
| Cov correction                              | +0.0023       |
| $\beta_{\text{formula}}$                    | <b>2.2571</b> |
| $\beta_{\text{actual}}$ (transfer events)   | <b>2.2571</b> |
| Residual                                    | <b>0.0000</b> |

The formula is exact. The covariance correction is small (+0.001). The main term dominates. All gap is in  $H_{\text{after}}$ .

## 2 Derived Components

$H_{\text{uniform}}$  and  $H_{\text{peaked}}$  from  $(\varepsilon_0, n)$

Born resampling:  $P_i \sim \text{Dir}(\alpha_d)$ ,  $\alpha_{d,k} = (\bar{P}_k + \varepsilon_0)/\varepsilon_0$ .

**Phase I neighbours (uniform):**  $\alpha_k = 1/(n\varepsilon_0) + 1 = 17.55$ ,  $H_{\text{uniform}} = \psi(n\alpha_k + 1) - \psi(\alpha_k + 1) = 2.055$

**Phase II neighbours (peaked):**  $\alpha_{\text{peak}} = 1/\varepsilon_0 = 132.5$ ,  $\alpha_{\text{bg}} = 1/n + 1 = 1.125$ ,  $H_{\text{peaked}} = 0.304$

Both derived from  $(\varepsilon_0, n)$  only. No free parameters.

## 3 The $f_{II} = 1/n$ Hypothesis

**Hypothesis:**  $f_{II} = 1/n$  from  $\Psi$ -Mode Symmetry

**Argument:**

1. V20.3 produces  $n = 8$  distinct  $\Psi$ -modes (observed; connected to  $\text{SO}(3) \rightarrow n = 8$ ).
2. In steady state, modes are equally populated by symmetry: each holds fraction  $1/n$  of nodes.
3. When node  $i$  fires, its neighbours that share the same  $\Psi$ -mode have peaked  $\bar{P}$ .
4. Same-mode fraction =  $1/n$  (one mode out of  $n$  equally populated modes).
5. Therefore:  $f_{II} = 1/n$ .

$$f_{II} = \frac{1}{n} = \frac{1}{8} = 0.125 \quad (\text{derived from } \text{SO}(3) \rightarrow n)$$

$$H_{\text{after}} = \left(1 - \frac{1}{n}\right) H_{\text{uniform}} + \frac{1}{n} H_{\text{peaked}} = \frac{7}{8}(2.055) + \frac{1}{8}(0.304) = 1.836$$

## 4 Remaining Gap

16% Gap in  $\beta$ :  $f_{II}^{\text{derived}}$  vs  $f_{II}^{\text{measured}}$

|                          | Derived       | Measured |
|--------------------------|---------------|----------|
| $f_{II}$                 | $1/n = 0.125$ | 0.147    |
| $H_{\text{after}}$       | 1.836         | 1.798    |
| $\beta$ (from main term) | 2.617         | 2.257    |
| Gap                      | 16%           |          |

The 16% gap maps exactly to the difference  $f_{II}^{\text{measured}} - 1/n = 0.147 - 0.125 = 0.022$ .

**Why  $f_{II} > 1/n$ ?** The equal-population argument assumes  $\Psi$ -modes are perfectly symmetric. In practice, nodes with high  $\Omega$  (many R-events) preferentially occupy specific  $\Psi$ -modes and fire more often — slightly enhancing the fraction of peaked-neighbour events beyond  $1/n$ . The enhancement factor  $e = f_{II}/(1/n) = 0.147/0.125 = 1.18$ .

**Deriving  $e$  from the  $\delta$ -chain:**  $e$  depends on the correlation between  $\Omega_i$  and  $\Psi$ -mode occupancy. This connects to HN-03 (collective  $\Omega$  excess in clusters). If high- $\Omega$  nodes preferentially occupy the same  $\Psi$ -mode, their firing rate is enhanced, increasing  $f_{II}$  above  $1/n$ .

## 5 Status

| Component              | Status            | Value / Source                        |
|------------------------|-------------------|---------------------------------------|
| $\varepsilon_0$        | Derived           | EIC/SINDy                             |
| $n$                    | Derived           | SO(3) #16                             |
| $H_{\text{uniform}}$   | Derived           | $\psi$ -formula, $(\varepsilon_0, n)$ |
| $H_{\text{peaked}}$    | Derived           | $\psi$ -formula, $(\varepsilon_0, n)$ |
| Formula exactness      | Verified          | Residual = 0.000                      |
| $f_{II} = 1/n$         | Hypothesis        | From $\Psi$ -mode symmetry            |
| Enhancement $e = 1.18$ | Open              | Requires HN-03                        |
| $\beta$ at 16%         | Hypothesis<br>gap | Closes with $e$                       |
| <b>#21 formal</b>      | <b>Open</b>       | <b>at Reduces from prev. 41%</b>      |
|                        | <b>16%</b>        |                                       |

*The formula is exact. The covariance terms cancel almost completely.*

*All gap is in  $H_{\text{after}}$ , which is in  $f_{II}$ , which is in  $e$ .*

*$f_{II} = 1/n$  is derivable from SO(3)  $\Psi$ -mode symmetry.*

*The correction  $e = 1.18$  connects to HN-03:  $\Omega$ -correlated cluster firing.*

*The chain from  $\delta$  to  $\beta$  is 84% derived.*

*The remaining 16% has a name:  $\Psi$ -mode firing enhancement.*